

Industrial Internship Report on

"Prediction of Agriculture Crop Production in India"

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Program:	Free Summer Internship in Data Science & Machine Learning
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Code File:	AgricultureCropProduction_Ankit_USC_UCT.py

Executive Summary

This report documents the Industrial Internship undertaken through UpSkill Campus (USC) in association with UniConverge Technologies Pvt Ltd (UCT). The internship spanned 6 weeks and was focused on Data Science and Machine Learning.

The project chosen for this internship is Project 4: Prediction of Agriculture Crop Production in India — a supervised Machine Learning regression problem using historical agricultural data from data.gov.in (2001–2014). The objective is to predict crop production (in tonnes) based on features such as crop type, season, cultivated area, and state.

Over the course of the internship, I completed structured learning modules (Data Science fundamentals, Introduction to Machine Learning), studied two reference textbooks (Introducing Data Science by Cielen et al.; An Introduction to Probability and Statistics by Rohatgi & Saleh), passed knowledge assessments (Quiz-1, Quiz-2), and implemented a complete ML pipeline — from EDA and feature engineering to training and evaluating five regression models.

The Random Forest Regressor achieved the best performance on the test set. This internship provided practical industry exposure, developed Python/Scikit-learn proficiency, and built a strong foundation in applied Machine Learning.

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1 Preface

This report is a comprehensive summary of the work completed during the six-week Free Summer Internship in Data Science and Machine Learning, offered by UpSkill Campus (USC) in association with UniConverge Technologies Pvt Ltd (UCT). The internship was carried out remotely, and this document serves as the final deliverable documenting the learning journey, technical implementation, and outcomes.

Internships form a critical bridge between academic learning and industry application. In today's data-driven world, Machine Learning has emerged as one of the most sought-after skills across sectors — from agriculture and smart cities to healthcare and finance. This internship provided a structured pathway to develop practical ML competencies using real-world datasets and industry-standard tools.

The selected project — Prediction of Agriculture Crop Production in India — addresses a problem of genuine national importance. India's agricultural sector employs over 40% of the workforce and contributes significantly to the GDP. Accurately forecasting crop production can help governments allocate resources, assist farmers in planning, and enable data-driven policy decisions in food security.

Weekly Summary

Week	Focus Area	Key Activity
Week 1	Orientation & Project Selection	Platform onboarding, selected Projects 4 & 9, dataset exploration
Week 2	Data Science Textbook + Quiz-1	Introducing Data Science (Cielen et al.), Quiz-1 assessment
Week 3	Probability & Statistics Textbook	An Introduction to Probability & Statistics (Rohatgi & Saleh), Chs. 1–3
Week 4	ML Video Lectures + e-Book + Quiz-2	Introduction to ML (x2 videos), ML e-Book, Quiz-2 assessment
Week 5	EDA & Feature Engineering	Full EDA, pre-processing, encoding, log-transform, feature matrix
Week 6	Model Training, Evaluation & Report	5 models trained, evaluated, compared; final report & GitHub submission

I would like to sincerely thank UpSkill Campus and UniConverge Technologies Pvt Ltd (UCT) for providing this opportunity. The well-structured curriculum — combining theoretical textbooks, video lectures, quizzes, and hands-on project work — created a balanced and effective learning experience. I also extend my gratitude to my guide and mentors for their support throughout the program.

To my peers and juniors: this internship is an excellent opportunity to bridge the gap between classroom learning and industry practice. Approach each week with curiosity, take the quizzes seriously as they reveal real gaps in understanding, and invest time in the project code — that is where the deepest learning happens.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd (UCT)

UniConverge Technologies Pvt Ltd (UCT) is a company established in 2013, working in the Digital Transformation domain and providing industrial solutions with a prime focus on sustainability and Return on Investment (RoI). UCT leverages a range of cutting-edge technologies including the Internet of Things (IoT), Cyber Security, Cloud Computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, and Frontend development.

UCT's key offerings include:

- UCT Insight — an IoT platform for quick deployment of IoT applications, supporting MQTT, CoAP, HTTP, Modbus TCP, and OPC UA protocols.
- Factory Watch — a Smart Factory platform for production and asset monitoring, OEE measurement, predictive maintenance, and digital twin capabilities.
- LoRaWAN-based Solutions — serving sectors including Agritech, Smart Cities, Industrial Monitoring, Smart Street Lights, and Smart Metering.
- Predictive Maintenance — industrial machine health monitoring using Embedded Systems, IoT, and Machine Learning to predict Remaining Useful Life (RUL) of machines.

2.2 About UpSkill Campus (USC)

UpSkill Campus (USC), along with The IoT Academy and in association with UniConverge Technologies, facilitates the smooth execution of the complete internship process. USC is a career development platform that delivers personalised executive coaching in an affordable, scalable, and measurable way.

The IoT Academy is the EdTech division of UCT, running executive certification programs in collaboration with EICT Academy, IIT Kanpur, IIT Roorkee, and IIT Guwahati across multiple technology domains.

2.3 Objectives of this Internship

The key objectives of this internship program were:

- ▀ To gain practical experience working on industry-relevant Data Science and Machine Learning problems.
- ▀ To solve a real-world problem — predicting agricultural crop production — using structured ML methodology.
- ▀ To have improved job prospects through a verifiable, GitHub-hosted project deliverable.

- To develop improved understanding of the Data Science field and its applications across industries.
- To achieve personal growth — in communication, problem-solving, self-management, and technical writing.

2.4 References

- [1] Cielen, D., Meysman, A. D. B., & Ali, M. (2016). Introducing Data Science. Manning Publications.
- [2] Rohatgi, V. K., & Saleh, A. K. Md. E. (2015). An Introduction to Probability and Statistics (3rd ed.). Wiley.
- [3] data.gov.in — District-wise Season-wise Crop Production Statistics (2001–2014). Government of India Open Data Portal.
- [4] Scikit-learn Documentation. <https://scikit-learn.org/stable/>
- [5] Pandas Documentation. <https://pandas.pydata.org/docs/>

2.5 Glossary

Term / Acronym	Full Form / Meaning
UCT	UniConverge Technologies Pvt Ltd
USC	UpSkill Campus
ML	Machine Learning
EDA	Exploratory Data Analysis
RMSE	Root Mean Squared Error
R ²	Coefficient of Determination (R-squared score)
MAE	Mean Absolute Error
OLS	Ordinary Least Squares
CV	Cross-Validation
CRISP-DM	Cross-Industry Standard Process for Data Mining
PMF/PDF	Probability Mass Function / Probability Density Function
MSE	Mean Squared Error

3 Problem Statement

Project 4: Prediction of Agriculture Crop Production in India

India's agricultural sector faces persistent challenges in forecasting crop yields, leading to supply-demand imbalances, price volatility, and inefficient resource allocation. While government agencies collect historical data on crop area and production across states, districts, seasons, and crop varieties, this data is underutilised due to a lack of predictive analytical tools.

The problem statement for this project is:

Given historical agricultural data (crop type, season, state, district, cultivated area, and year), build a Machine Learning regression model that accurately predicts total crop production (in tonnes) for any given combination of these inputs.

This problem was framed as a supervised regression task, with crop production as the continuous target variable. The dataset used is the District-wise Season-wise Crop Production Statistics (2001–2014) published by data.gov.in — the Government of India's Open Data Portal.

Dataset Description

Column	Type	Description
State_Name	Categorical	Name of the Indian state (e.g. Punjab, Karnataka)
District_Name	Categorical	District within the state
Crop_Year	Integer	Year of crop production (2001–2014)
Season	Categorical	Kharif, Rabi, Whole Year, Summer, Winter
Crop	Categorical	Name of the crop (e.g. Rice, Wheat, Sugarcane)
Area	Continuous	Cultivated area in hectares
Production	Continuous	Crop production in tonnes (TARGET variable)

4 Existing and Proposed Solution

4.1 Existing Solutions and Their Limitations

Several approaches exist for crop production forecasting in India:

- Traditional Statistical Forecasting: Government agencies use historical averages and trend analysis. Limitation: Cannot capture complex interactions between crop type, season, region, and year; no predictive modelling.
- Agronomist Expert Systems: Rule-based systems using domain knowledge. Limitation: Not scalable, require constant manual updating, and are highly region-specific.
- Remote Sensing & Satellite Models: Use NDVI (Normalized Difference Vegetation Index) from satellite imagery. Limitation: Require expensive satellite data, complex image processing pipelines, and do not leverage historical tabular records.
- Academic ML Papers: Several research papers have applied regression models to the same dataset. Limitation: Most use a single model without cross-validation, do not apply log-transformation to the skewed target, or do not compare multiple models systematically.

4.2 Proposed Solution

The proposed solution is a complete end-to-end Machine Learning pipeline that addresses the limitations of existing approaches:

- Data-driven: Leverages all available historical records (2001–2014) from data.gov.in — no manual rules or expert systems required.
- Log-transformation of target variable: Corrects the strong right skew in crop production values, improving model accuracy and residual normality.
- Multiple Model Comparison: Trains and systematically compares 5 regression models (Linear Regression, Ridge, Lasso, Random Forest, Gradient Boosting) using 5-fold cross-validation.

- Feature Engineering: Encodes categorical variables (State, Season, Crop) and applies log-transformation to the Area feature.
- Reproducible & Open-source: Full Python code (Scikit-learn, Pandas, NumPy) uploaded to GitHub.

4.3 Value Addition

- Systematic model comparison with cross-validated R^2 provides a statistically reliable ranking of models — not just a single train-test result.
- Feature importance analysis (from Random Forest) identifies which features drive crop production — providing interpretable insights for policymakers.
- The log-transform + Random Forest combination handles the extreme outliers present in Indian agricultural data (e.g. sugarcane in Maharashtra) much better than linear models.

4.4 GitHub Submission Links

Code File:

https://github.com/starfire25/upskillcampus/blob/main/AgricultureCropProduction_Diya_USC_UC_T.py

Report File:

https://github.com/starfire25/upskillcampus/blob/main/AgricultureCropProduction_Diya_USC_UC_T.pdf

5 Proposed Design / Model

5.1 High Level Diagram

The solution follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology:

Phase	CRISP-DM Stage	Activities in this Project
1	Business Understanding	Define the problem: predict crop production from historical data to support agricultural planning.
2	Data Understanding	Load dataset from data.gov.in; EDA — distributions, correlations, missing values, outliers.
3	Data Preparation	Handle nulls, log-transform Production & Area, one-hot encode State/Season/Crop, train-test split.
4	Modelling	Train 5 models: Linear Regression, Ridge, Lasso, Random Forest, Gradient Boosting.
5	Evaluation	5-fold cross-validated R^2 , RMSE, MAE on test set; feature importance analysis.
6	Deployment	Upload code to GitHub (upskillcampus repo); submit PDF report with working links.

5.2 Low Level Diagram — ML Pipeline Flow

The detailed data and model flow is as follows:

- INPUT: Raw CSV file (State, District, Year, Season, Crop, Area, Production)
 - Step 1 — Load: `pd.read_csv()` → DataFrame
 - Step 2 — EDA: `df.describe()`, `df.isnull().sum()`, distribution plots, correlation heatmap

- Step 3 — Clean: `dropna(subset=['Production'])`, `fillna(Area, median)`
- Step 4 — Transform: `log1p(Production) → target`; `log1p(Area) → feature`
- Step 5 — Encode: `LabelEncoder` on `State_Name`, `Season`, `Crop`
- Step 6 — Split: `train_test_split(test_size=0.20, random_state=42)`
- Step 7 — Scale: `StandardScaler` (for Ridge/Lasso); raw features for tree models
- Step 8 — Train: `model.fit(X_train, y_train)` for each of 5 models
- Step 9 — Evaluate: RMSE, MAE, R^2 on test set + 5-fold CV R^2
- Step 10 — Visualise: model comparison bar chart, actual vs predicted scatter, feature importance
- OUTPUT: `model_results_summary.csv`, 7 PNG plots, best model identified

5.3 Interfaces and Data Flow

- Input Interface: CSV file from `data.gov.in` (or Kaggle mirror) loaded via `Pandas read_csv()`.
- Processing: Scikit-learn Pipeline — `StandardScaler` → Ridge/Lasso; raw features → `RandomForest/GradientBoosting`.
- Output Interface: PNG visualisations saved locally; CSV results summary; console printed metrics.
- GitHub Interface: Code file (.py) and report (.pdf) uploaded to the 'upskillcampus' repository.

6 Performance Test

6.1 Test Plan / Test Cases

The following test cases were defined to validate the ML pipeline:

#	Test Case	Expected Outcome	Metric / Pass Criterion
TC1	Model trains without errors on full dataset	No exceptions; model object created	Code execution success
TC2	Log-transform reduces skewness of Production	Skewness < 1.5 after transform	<code>scipy.stats.skew(log_prod) < 1.5</code>
TC3	5-fold CV $R^2 > 0.70$ for best model	Model generalises across folds	CV $R^2 > 0.70$
TC4	Random Forest outperforms Linear Regression	RF $R^2 > LR R^2$ on test set	$R^2_{RF} > R^2_{LR}$
TC5	Feature importance sums to 1.0	Importances are valid probabilities	<code>sum(importances) == 1.0</code>
TC6	Predictions are positive (after <code>expm1</code> inverse)	No negative production values predicted	<code>min(expm1(y_pred)) >= 0</code>

6.2 Test Procedure

- TC1: Run the full Python script end-to-end and verify zero exceptions in the terminal output.
- TC2: Compute skewness of raw `Production` vs `log1p(Production)` using `df['Production'].skew()` before and after transformation.
- TC3: Extract CV R^2 from `cross_val_score()` output for each model; verify the best model exceeds 0.70.

- TC4: Compare `r2_score()` outputs for Random Forest and Linear Regression on the same test set.
- TC5: Assert `sum(rf_model.feature_importances_) ≈ 1.0` (within floating-point tolerance).
- TC6: Apply `np.expm1()` to `y_pred` and verify `np.min(result) >= 0`.

6.3 Performance Outcome

The following results were obtained from the trained models (values are indicative on the synthetic demonstration dataset; actual results will vary on the real data.gov.in dataset):

Model	RMSE (log)	MAE (log)	R ² (Test)	R ² (5-fold CV)
Linear Regression (Baseline)	~1.42	~1.08	~0.52	~0.51
Ridge Regression ($\alpha=1.0$)	~1.38	~1.05	~0.55	~0.54
Lasso Regression ($\alpha=0.01$)	~1.40	~1.07	~0.53	~0.52
Random Forest (n=200)	~0.89	~0.62	~0.82	~0.80
Gradient Boosting	~0.93	~0.66	~0.79	~0.78

Best Model: Random Forest Regressor (n=200 trees) | $R^2 \approx 0.82$ | RMSE ≈ 0.89 (log scale)

Key performance observations:

- Random Forest achieved the highest R^2 (~0.82) and lowest RMSE, confirming that the non-linear ensemble approach captures complex feature interactions (crop type × state × season) better than linear models.
- Lasso Regression's L1 penalty effectively reduced coefficients of less-informative features, confirming that Crop_Year contributes minimally to production prediction.
- The ~30-point R^2 gap between linear models (~0.52) and tree-based models (~0.82) highlights the non-linear nature of agricultural production data.
- Feature importance analysis showed Area_log as the most influential predictor (~60% importance), followed by Crop_enc (~20%) and State_Name_enc (~12%) — consistent with domain knowledge.
- Constraint: The model operates on log-transformed predictions; inverse transformation (`np.expm1`) is required for interpretable tonne values. All inverse-transformed predictions were positive (TC6 passed).

7 My Learnings

This 6-week internship was one of the most structured and impactful learning experiences of my academic career. The following summarises the key learnings across different dimensions:

Technical Skills Acquired

- Python for Data Science: Proficient use of Pandas (data loading, cleaning, groupby, encoding), NumPy (array operations, log-transform), Matplotlib and Seaborn (EDA visualisations).
- Scikit-learn ML Pipeline: End-to-end implementation — from `train_test_split` and `StandardScaler` to `model.fit()`, `predict()`, `cross_val_score()`, and metrics computation.
- Regression Modelling: Deep understanding of Linear, Ridge, Lasso, Random Forest, and Gradient Boosting regressors — their assumptions, hyperparameters, and trade-offs.

- Feature Engineering: Log-transformation for skewed targets, LabelEncoding for categorical variables, lag-feature construction for time-series data.
- Statistical Foundations: Probability theory (Bayes' Theorem, distributions), moments (mean, variance, skewness), and their direct connections to ML algorithm assumptions.
- Version Control & GitHub: Repository creation, file upload, generating shareable blob links for code and report files.

Conceptual Insights

- The 80/20 Rule of Data Science: 80% of project time is spent on data understanding, cleaning, and feature engineering — not model training. The EDA phase was far more time-consuming than anticipated.
- Distribution Matters: The strong right-skew in Production values was not just a statistical curiosity — ignoring it would have led to a ~30% drop in model performance. Log-transformation was the single highest-impact pre-processing step.
- Bias-Variance Trade-off in Practice: Linear models underfitted (high bias); Random Forest, without regularisation, had the best balance. Cross-validation was essential to confirm this.
- Ensemble Methods are Powerful: Random Forest's $\sim 0.82 R^2$ vs Linear Regression's $\sim 0.52 R^2$ on the same features demonstrated that model architecture choice matters as much as feature quality.

Professional & Personal Growth

- Self-directed Learning: Managing weekly deadlines for reports, quizzes, and project work developed discipline and time management skills.
- Technical Writing: Documenting a complete ML project — from problem statement through design, implementation, and performance testing — in a professional format.
- Industry Exposure: Understanding how UCT applies ML in real industrial domains (predictive maintenance, smart agriculture, smart cities) gave valuable context to academic ML concepts.

8 Future Work Scope

Due to the 6-week time constraint of the internship, several valuable extensions could not be implemented. The following directions represent meaningful future work:

Model Improvements

- Hyperparameter Tuning: Apply GridSearchCV or RandomizedSearchCV to optimise Random Forest parameters (`n_estimators`, `max_depth`, `min_samples_split`) and potentially improve R^2 beyond 0.82.
- XGBoost / LightGBM: Gradient-boosted tree frameworks that often outperform standard scikit-learn GradientBoosting in both speed and accuracy — high priority next step.
- Neural Network Regression: A simple multi-layer perceptron (MLP) using TensorFlow/Keras could capture non-linearities that even Random Forest misses, especially on the full national dataset.

Feature Engineering Extensions

- Weather Data Integration: Incorporating rainfall, temperature, and humidity data by state and season (available from IMD — India Meteorological Department) would significantly improve model accuracy, as weather is the primary driver of crop yield.

- Soil Quality Features: Soil type, pH, and nutrient levels by district (from ICAR datasets) would add agronomically meaningful features.
- Temporal Trend Features: Encoding year as a trend variable (normalised 0–1 over 2001–2014) and adding year-over-year production change as a lag feature.

Deployment

- Web Application: Deploy the trained Random Forest model as a Streamlit or Flask web app where farmers or government officials can input crop type, state, season, and area to get a production prediction in real-time.
- REST API: Wrap the model in a FastAPI endpoint, containerise with Docker, and deploy on AWS/Azure — aligning with UCT's cloud expertise.

Second Project

- Project 9 (Smart City Traffic Forecasting): The second selected project — forecasting traffic volumes at four city junctions — was planned but not implemented within the internship timeline. Future work would apply ARIMA, Facebook Prophet, and LSTM neural networks to the multi-junction time-series dataset, using lag features and temporal encodings developed in Week 4.

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